

Comparitive Performance Report for direct-reads-pr2-3-way-replica-balance- reads-off-rerun2 vs direct-reads-pr2-3-way-replica-balance- reads-on-split-256k-rerun2 vs tentacle-3-way-replica-balance-reads-off

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Comparison summary for direct-reads-pr2-3-way-replica-balance- reads-off-rerun2 vs direct-reads-pr2-3-way-replica-balance- reads-on-split-256k-rerun2 vs tentacle-3-way-replica-balance-reads-off

Sequential Read	direct_reads_pr2_3way_replica_balance_reads_off_rerun2	direct_reads_pr2_3way_replica_balance_reads_on_split_256k_rerun2	diff	tentacle_3way_replica_balance_reads_off	diff
4K	158346@2.4ms	264393@1.4	67%	157548@2.0	-1%
8K	151639@2.5ms	258173@1.5	70%	150852@2.5	-1%
16K	139269@2.8ms	222987@1.7	60%	138108@2.8	-1%
32K	122178@3.1ms	194498@2.0	59%	121533@3.2	-1%
64K	6385@3.9ms	9403@2.7	47%	6331@4.0	-1%
128K	10103@5.0ms	13721@3.7	36%	9823@5.1	-3%
256K	14785@6.8ms	17736@5.7	20%	14295@7.0	-3%
384K	15830@12.7ms	17239@5.8	9%	14995@6.7	-5%
512K	16707@8.0ms	14796@9.1	-11%	17061@15.7	2%
768K	15548@8.1ms	15673@8.0	1%	17212@23.4	11%
1024K	15688@8.5ms	15290@4.4	-3%	16689@24.1	6%
2048K	13053@10.3ms	16067@8.3	23%	14689@9.1	13%
4096K	12305@10.9ms	14568@9.2	18%	13597@59.2	10%

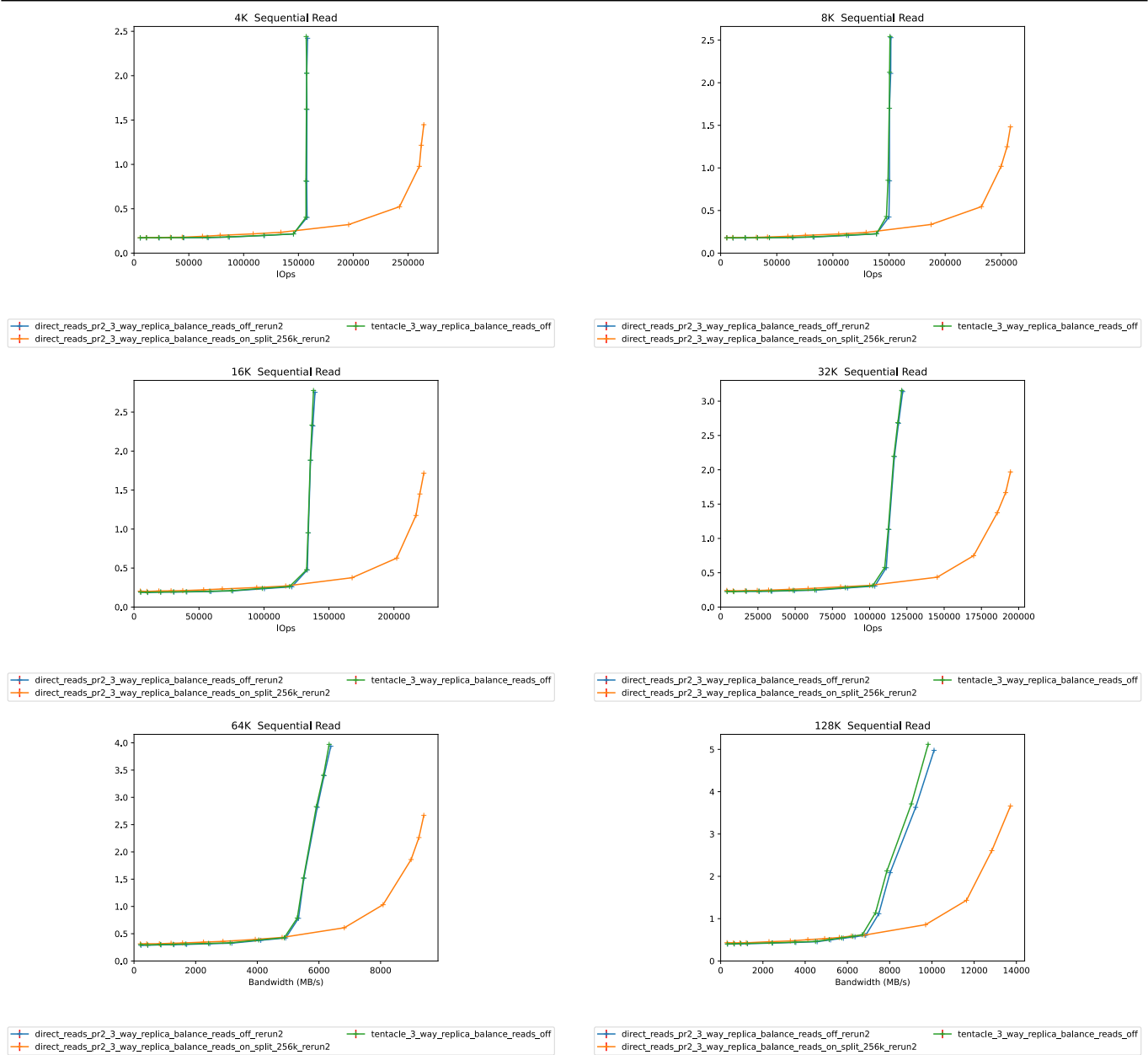
Random Read	direct_reads_pr2_3way_replica_balance_reads_off_rerun2	direct_reads_pr2_3way_replica_balance_reads_on_split_256k_rerun2	diff	tentacle_3way_replica_balance_reads_off	diff
4K	233097@1.6ms	250356@1.3	7%	219491@1.2	-6%
8K	228795@1.7ms	230551@1.1	1%	209044@1.2	-9%
16K	203900@1.9ms	208560@1.5	2%	188007@1.4	-8%
32K	181336@1.8ms	189493@1.3	4%	167762@2.3	-7%
64K	9603@1.7ms	9699@2.6	1%	9032@2.8	-6%
128K	14375@3.5ms	13922@2.4	-3%	13244@3.8	-8%
256K	18121@3.7ms	17190@5.8	-5%	16481@4.1	-9%
384K	17080@2.9ms	17698@11.4	4%	17128@11.7	0%
512K	17128@3.9ms	15681@17.1	-8%	17600@15.2	3%
768K	16397@3.1ms	15858@3.2	-3%	17110@23.5	4%
1024K	14922@9.0ms	15949@25.2	7%	16507@16.2	11%
2048K	13509@9.9ms	16031@8.3	19%	14724@18.2	9%
4096K	12830@10.4ms	14976@17.9	17%	13969@57.6	9%

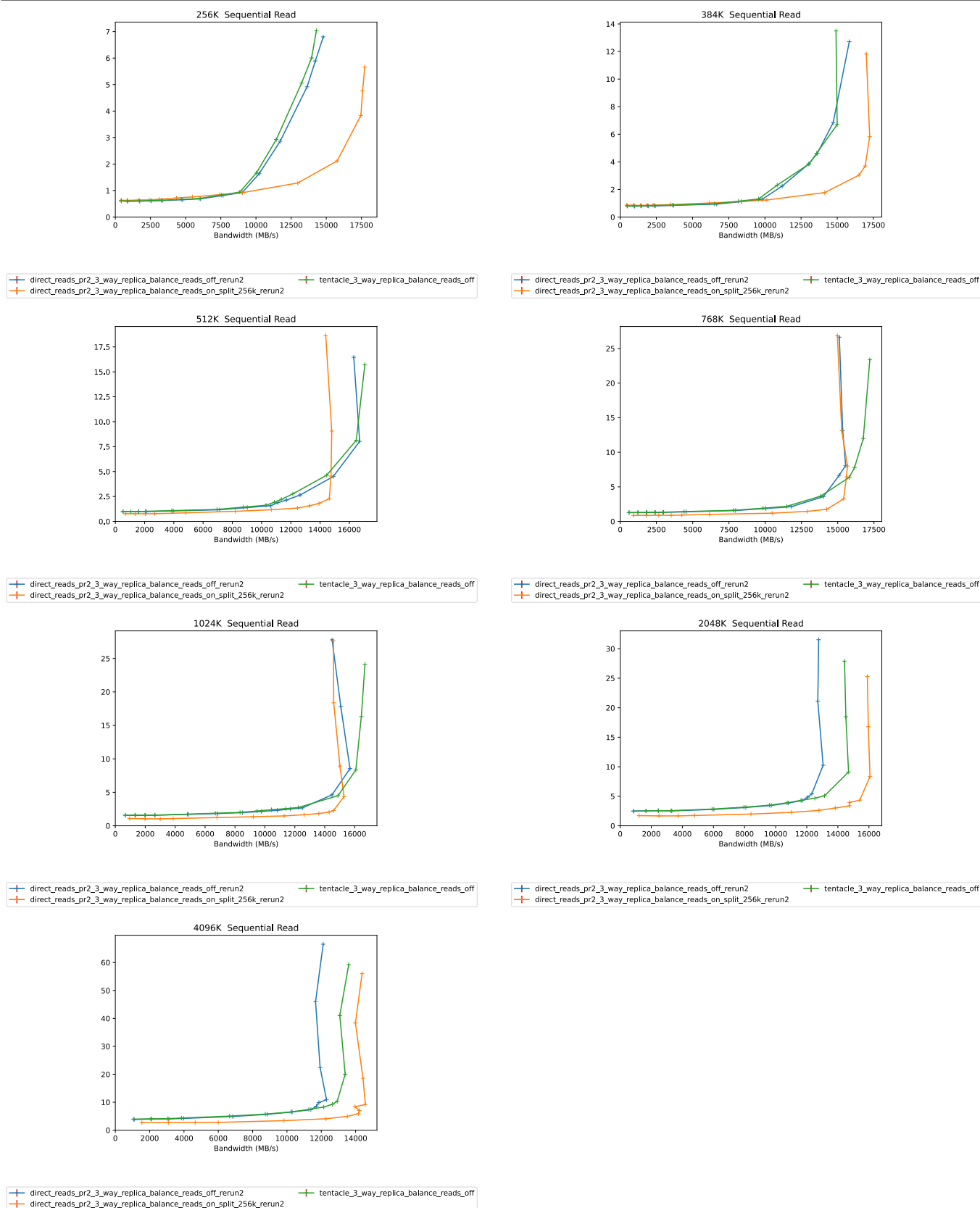
Random Read/Write	direct_reads_pr2_3way_replica_balance_reads_off_rerun2	direct_reads_pr2_3way_replica_balance_reads_on_split_256k_rerun2	diff	tentacle_3way_replica_balance_reads_off	diff
4K_70/30	121693@3.1ms	129777@3.0	7%	119218@3.2	-2%
16K_70/30	105266@3.6ms	108980@3.5	4%	101882@3.8	-3%

Random					
Read/Write	direct_reads_pr2_3way_replica_balance_reads_off_rerun2	direct_reads_pr2_3way_replica_balance_reads_on_split_256k_rerun2	diff_balance	tentacle_3way_replica_balance_reads_off	diff_balance
64K_30/70	3184@5.2ms	3182@7.9	-0%	3170@5.3	-0%
64K_70/30	5084@4.9ms	5112@4.9	1%	4847@4.3	-5%

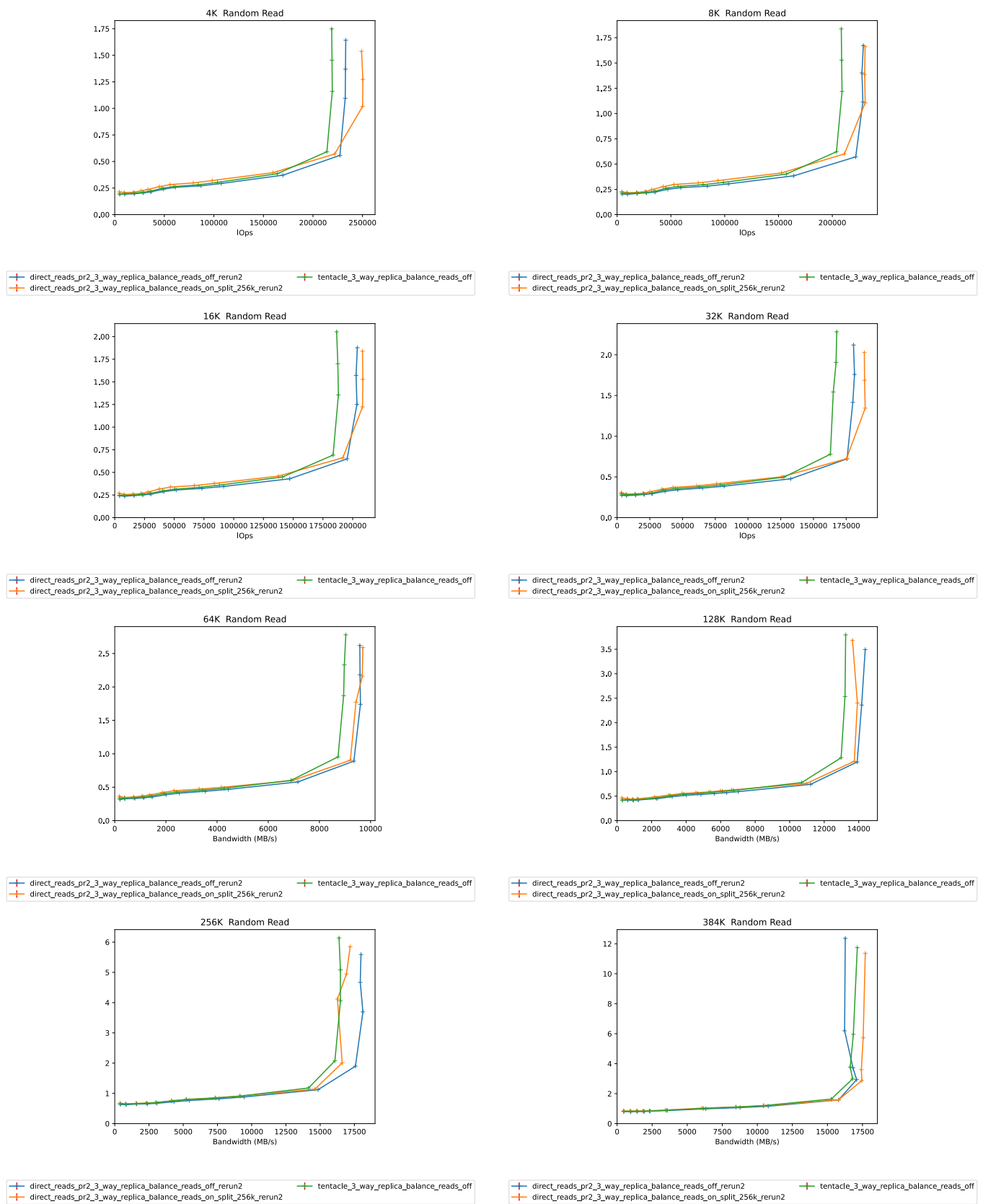
Response Curves

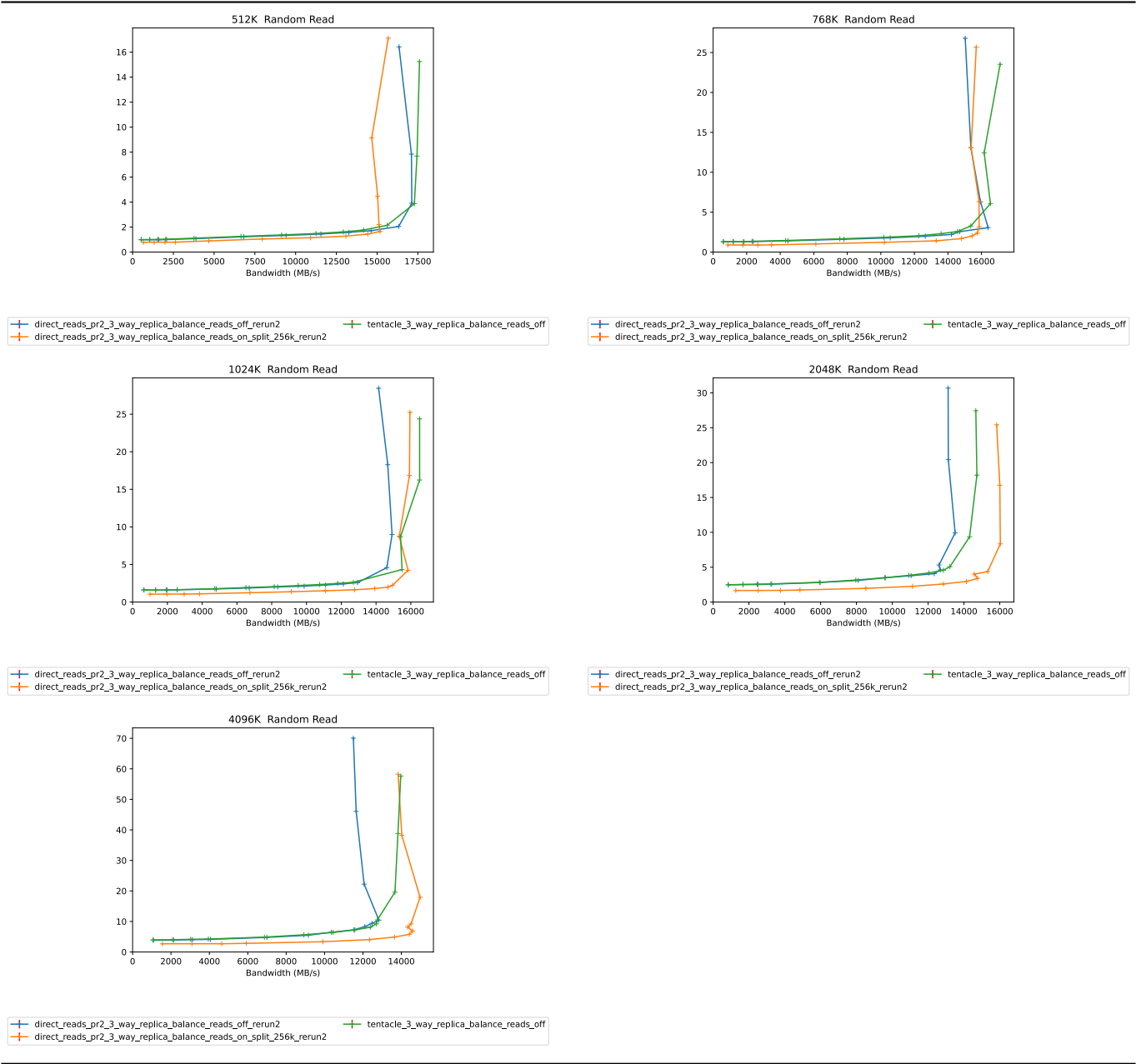
Sequential Read



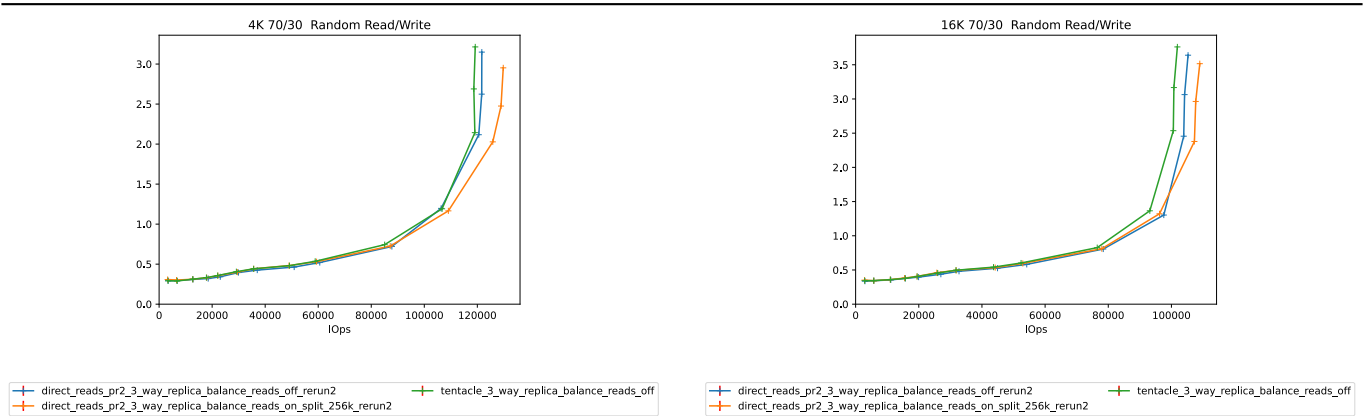


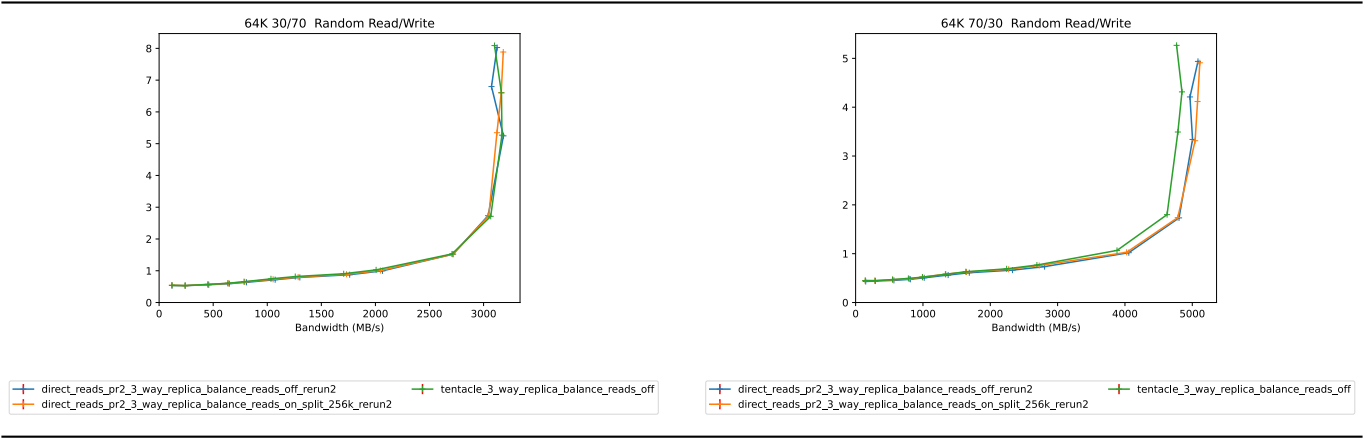
Random Read





Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```
librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
```

```
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
```

```
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
```

```
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
```

```
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
```

```
- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
```


- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
 - 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
 - 1
 - 2
 - 4
 - 6
 - 8
 - 12
 - 16
 - 24
 - 32
 - 64
 - 128
 - 256
 - 320
 - 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
 - 1
- op_size: 4096
- total_iodepth:
 - 1
 - 2
 - 4
 - 6
 - 8
 - 12
 - 16
 - 24
 - 32
 - 64
 - 128
 - 256
 - 320
 - 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
 - 1
- op_size: 4096
- total_iodepth:

```
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30
```

```
total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
```

```
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```